

AI Legal / Tax Research Assistant

Complete Project Documentation

Project Name	AI Legal / Tax Research Assistant
Contest	Indian Servers AI Model Development Contest 2026
Problem Statement	#03 — AI Legal / Tax Research Assistant
Developer	Isaac Gumpula
Live URL	https://legal-ai-assistant-omega.vercel.app
Tech Stack	LLaMA 3.3 70B · Groq API · RAG · PDF.js · Vercel
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■■ This tool is for research and educational support only. It is NOT a substitute for qualified legal or tax advice.

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1. Project Overview

The AI Legal / Tax Research Assistant is a web-based application that empowers users to interact with legal and tax documents using natural language. Instead of reading through lengthy PDFs manually, users can upload any document and instantly ask questions, generate compliance checklists, detect risk flags, and compare two documents side by side — all powered by state-of-the-art AI.

Built for the Indian Servers AI Model Development Contest 2026, this project addresses the real-world challenge of making complex legal and tax documents accessible and understandable to everyone — from students and small business owners to compliance officers and legal researchers.

2. Problem Statement

Legal and tax documents are notoriously complex, lengthy, and filled with jargon that is difficult for non-experts to understand. Key challenges include:

- Time-consuming manual review of lengthy documents (GST rules, IT Act, compliance docs)
- Difficulty identifying relevant clauses, deadlines, and obligations
- High cost of consulting lawyers or tax professionals for basic queries
- Risk of missing critical compliance requirements or penalty clauses
- No easy way to compare old vs new provisions in updated regulations

This application solves all the above by providing an AI-powered document assistant that gives instant, cited answers with section references — making legal research fast, affordable, and accessible.

3. Tech Stack

Category	Technology	Purpose
AI Model	LLaMA 3.3 70B	Large language model for answering questions
AI API	Groq API	Ultra-fast inference (500+ tokens/sec)
Architecture	RAG (Retrieval-Augmented Generation)	Context-aware document Q&A
PDF Processing	PDF.js (v3.11)	Extract text from uploaded PDFs in browser
Frontend	HTML5 + CSS3 + JavaScript (ES6+)	UI, chat interface, interactions
Backend	Node.js Serverless Function	Secure API proxy on Vercel
Deployment	Vercel	Hosting, CI/CD, environment variables
Version Control	Git + GitHub	Source code management
Voice Input	Web Speech API	Browser-native speech recognition
Storage	localStorage (browser)	Session memory and chat persistence

4. System Architecture & Flow

4.1 RAG Architecture Explained

RAG (Retrieval-Augmented Generation) is the core architecture of this application. Instead of sending the entire document to the AI (which is expensive and slow), the system intelligently retrieves only the most relevant sections before generating an answer.

4.2 Complete Application Flow

Step	Action	Description
1	PDF Upload	User uploads a PDF file via the browser interface
2	Text Extraction	PDF.js reads the PDF and extracts all text content client-side
3	Chunking	Text is split into overlapping chunks (~500 chars each) for RAG
4	User Query	User types or speaks a question in the chat interface
5	Retrieval	Most relevant chunks are retrieved based on keyword matching
6	Prompt Building	System prompt + relevant chunks + chat history are combined
7	API Call	Request sent to /api/groq (serverless function) — API key stays hidden
8	Groq Processing	LLaMA 3.3 70B processes the prompt via Groq's fast inference
9	Response	AI answer returned with section references and citations
10	Render	Response displayed in chat UI with formatted markdown
11	Memory	Chat + document saved to localStorage for session persistence

5. Features

■ Document Upload & Processing

Upload any legal, tax, or compliance PDF. The app extracts text using PDF.js directly in the browser — no file is sent to any server. Supports documents like GST rules, Income Tax Act, IFRS standards, audit reports, contracts, and more.

■ Ask Your Document (Q&A; with Citations)

Type any question about the uploaded document and get an AI-generated answer with references to specific sections, clauses, and articles. The RAG system ensures answers are grounded in the actual document content.

■ Auto Document Summary

One-click summarization of the entire document. The AI generates a structured summary covering key obligations, important provisions, parties involved, and critical dates.

■ Compliance Checklist Generator

Automatically generates an actionable compliance checklist from the document. Each checklist item is derived from the document's requirements — useful for audit preparation and regulatory compliance.

■ Risk Flag Scanner

Scans the document for potential risk clauses, penalty provisions, ambiguous language, and high-risk obligations. Each flag includes an explanation of why it is flagged and relevant section references.

■ Document Comparison

Upload two documents (e.g., old vs new version of a regulation) and compare them side by side. The AI highlights key differences, new provisions, removed clauses, and changed obligations.

■ Export Full Report

Export the entire chat session as a formatted HTML report or plain .txt file. The HTML report includes all questions, AI answers, citations, and timestamps — suitable for sharing with teams.

■ Voice Input

Ask questions using your microphone via the Web Speech API. Useful for hands-free operation while reviewing documents on screen.

■ Session Memory & Chat Persistence

Chat history and the uploaded document are saved to localStorage. On returning to the app, the previous session is fully restored — users can continue asking questions without re-uploading the document.

■ Sample Questions

Pre-loaded sample questions help users get started quickly: 'What are the key obligations?', 'What penalties are mentioned?', 'What are the filing deadlines?', etc.

■ Mobile-Friendly UI

Fully responsive design works on phones, tablets, and desktops. The chat interface adapts to all screen sizes.

6. Advantages

#	Advantage	Details
1	Speed	Get answers from a 100-page document in under 5 seconds — vs hours of manual reading
2	Cost-Effective	Free to use; reduces need for expensive legal consultations for basic queries
3	Privacy-First	PDFs are processed entirely in the browser — document text never leaves the user's device
4	RAG Accuracy	Answers are grounded in the actual document, not general AI hallucinations
5	Section References	Every answer includes references to specific clauses and sections
6	Multi-Use	Works for GST, Income Tax, IFRS, company law, contracts, audit standards and more
7	No Installation	100% web-based — works in any modern browser, no software to install
8	Fast Inference	Groq API provides ~500 tokens/second — near-instant AI responses
9	Session Persistence	No need to re-upload documents — session is restored automatically
10	Export Ready	Generated reports can be downloaded and shared with colleagues or clients
11	Secure API	API key stored as server environment variable — never exposed in frontend code
12	Dual Export	Export as formatted HTML report or plain text — flexible for different use cases

7. Disadvantages / Limitations

#	Limitation	Details
1	Not a Legal Opinion	The tool provides research support only — it cannot replace a qualified lawyer or CA
2	Scanned PDFs	If a PDF is scanned (image-based), PDF.js cannot extract text — OCR not implemented yet
3	Context Window Limit	Very large documents (500+ pages) may exceed the LLM context window
4	Language Support	Currently English only — Hindi, Telugu, and regional languages not yet supported
5	No Cloud Storage	Documents stored only in browser localStorage — clearing browser data loses history
6	Internet Required	Requires internet connection for Groq API calls — no offline mode
7	Groq Rate Limits	Free Groq API tier has rate limits — heavy usage may hit limits
8	Hallucination Risk	Like all LLMs, may occasionally generate incorrect information — always verify
9	No User Authentication	No login system — all users share the same public deployment
10	Table Extraction	Complex tables in PDFs may not be extracted with perfect formatting

8. Upcoming Features

■ Multi-language Support

Telugu, Hindi, Tamil, and other regional language support for wider accessibility

■ Multi-Document Search

Search and query across multiple uploaded documents simultaneously

■ Visual Analytics

Charts showing clause distribution, risk scores, and compliance percentages

■ User Authentication

Login system with personal document history and saved sessions

■ Email Summary

Send generated summaries and reports directly via email from within the app

■ Deadline Alerts

Extract and set reminders for filing deadlines and compliance dates

■ Mobile App

Native Android/iOS app for on-the-go document analysis

■ Fine-tuned Model

Domain-specific fine-tuned model on Indian legal and tax corpus

■■ Cloud Storage

Save documents and chat history to cloud for cross-device access

■■ OCR Support

Extract text from scanned/image-based PDFs using OCR technology

■ Template Library

Pre-built templates for common documents — GST returns, IT forms, etc.

■ Version Tracking

Track changes across multiple versions of the same regulation over time

9. User Manual — How to Use

9.1 Getting Started

Open the app at: <https://legal-ai-assistant-omega.vercel.app>

No login, no installation required. Works on any modern browser (Chrome recommended).

9.2 Step-by-Step Usage Guide

Step 1: Upload a Document

- Click the 'Click or drag & drop PDF' area on the left sidebar
- Select any legal or tax PDF from your computer
- Wait 2-5 seconds for text extraction to complete
- You will see 'X sections extracted' confirmation
- Quick Action buttons (Summarize, Checklist, Risk Flags) will become active

Step 2: Ask Questions

- Type your question in the chat box at the bottom
- Press Enter to send (Shift+Enter for new line)
- OR click the  microphone button to speak your question
- The AI will respond with an answer referencing specific sections
- You can ask multiple follow-up questions — the AI remembers context

Step 3: Use Quick Actions

- Click ' Summarize Document' for a full document summary
- Click ' Generate Compliance Checklist' for actionable checklist
- Click ' Scan for Risk Flags' to identify risky clauses

Step 4: Compare Two Documents

- After uploading Document 1, the 'Compare with Document 2' section appears
- Upload the second PDF (e.g., updated regulation)
- Click ' Compare Both Documents'
- The AI will highlight key differences between the two documents

Step 5: Export Your Report

- Click ' Export Full Report' for a formatted HTML report
- Click ' Export as .txt' for plain text export
- Click ' Clear Chat' to start a fresh session

Step 6: Sample Questions

- Click any question in the 'Try These Questions' section to auto-fill it
- Press Enter to send — great for getting started quickly

9.3 Tips for Best Results

- Use text-based PDFs (not scanned images) for best text extraction
- Ask specific questions like 'What is the penalty for late GST filing?' rather than 'Tell me about this'
- For large documents, use Quick Actions first to get an overview before asking specific questions
- The AI remembers your conversation — ask follow-up questions naturally
- If an answer seems incomplete, ask 'Can you elaborate on section X?'
- Use the Compare feature for budget documents, old vs new tax rules, or contract amendments

10. Deployment Guide

10.1 Project Structure

```
legal-ai-assistant/ ■■■ index.html ← Main UI (single page app) ■■■ style.css ←
All styling ■■■ app.js ← Frontend logic, RAG, PDF processing ■■■ vercel.json ←
Vercel deployment configuration ■■■ .gitignore ← Excludes .env from git ■■■ api/
■■■ groq.js ← Serverless function (API proxy)
```

10.2 Environment Variables

The following environment variable must be set in Vercel:

Variable	Value	Where to Set
GROQ_API_KEY	gsk_XXXXXXXXXXXXXXXXXXXX	Vercel → Settings → Environment Variables

10.3 Deployment Steps (Summary)

1. Install Git from git-scm.com
2. Create GitHub account and new repository
3. Run: git init, git add ., git commit -m 'Initial commit'
4. Run: git remote add origin
5. Run: git push -u origin main
6. Go to vercel.com → Sign up with GitHub
7. Import your GitHub repository
8. Add GROQ_API_KEY in Environment Variables
9. Click Deploy — your app is live in ~1 minute!

11. API & Model Details

11.1 LLaMA 3.3 70B

Property	Value
Model Name	LLaMA 3.3 70B Versatile
Parameters	70 Billion
Developer	Meta AI
Context Window	128,000 tokens
Inference Speed (Groq)	~500 tokens/second
Max Output Tokens	1024 (configured)
Temperature	0.3 (factual/precise mode)
Use Case	Document Q&A, summarization, legal analysis

11.2 Groq API

Groq provides LPU (Language Processing Unit) hardware specifically designed for LLM inference. This results in inference speeds 10-100x faster than traditional GPU-based providers, making the app feel near-instant to users. The API is OpenAI-compatible, making integration straightforward.

11.3 RAG Implementation

Chunking: Document text split into ~500 character chunks with overlap for context continuity

Retrieval: On each query, top-N most relevant chunks selected using keyword frequency scoring

Context Building: Selected chunks combined with system prompt and chat history

Generation: LLaMA 3.3 generates answer grounded in retrieved context

Citation: Model instructed via system prompt to cite section numbers in responses

12. Security & Privacy

- **API Key Security:** Groq API key stored as Vercel environment variable — never exposed in frontend code or GitHub
- **Document Privacy:** PDFs processed entirely in the user's browser using PDF.js — document content never uploaded to any server
- **CORS Protection:** API route includes CORS headers — only configured origins can call the backend
- **No Data Collection:** No user data, documents, or queries are stored on any server
- **HTTPS:** Vercel automatically provides HTTPS/SSL — all traffic encrypted in transit
- **Disclaimer:** App clearly states it is for research support only — not legal/tax advice

13. Evaluation Criteria Mapping

How this project addresses each evaluation criterion from the contest:

Criterion	Max Marks	How This Project Addresses It
Innovation	20	RAG architecture + voice input + dual doc comparison + session memory — beyond basic
AI/Model Implementation	20	LLaMA 3.3 70B via Groq API, RAG chunking/retrieval, structured prompting with citations
UI/UX & Interactivity	15	Clean sidebar+chat layout, quick actions, sample questions, voice input, mobile-friendly
Practical Usefulness	15	Solves real problem — instant answers from legal docs, compliance checklist, risk detection
Technical Architecture	10	Serverless backend, RAG pipeline, secure API proxy, localStorage persistence
Accuracy & Safety	10	Low temp (0.3), section citations, clear disclaimer, API key never exposed
Presentation Quality	10	Clear demo flow, well-structured UI, export feature for report sharing
TOTAL	100	Project addresses all 7 evaluation criteria comprehensively

Document prepared for Indian Servers AI Model Development Contest 2026

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